MODULE 5 CRANIAL NERVES

The Cranial Nerves.

THE PERIPHERAL NERVOUS SYSTEM, THE CRANIAL NERVES

A reference for the cranial nerve video and lab. This page covers the twelve pairs of cranial nerves: the number and name of each, its classification as sensory, motor, or mixed, its function, the skull foramen it passes through, and its nuclei. The focus is on the structures and the job each one does.

BY THE END

1. Name the twelve cranial nerves in order and give the number of each.
2. Classify each cranial nerve as sensory, motor, or mixed.
3. Identify the function, skull foramen, and nuclei of each cranial nerve.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The cranial nerves

drsrennie-stack.github.io/new-build-bio4-solano/cranial-nerve-videos.html

The Cranial Nerves, an Overview

Twelve pairs of cranial nerves arise from the brain. They are numbered I to XII from front to back by where they attach. The first two attach to the cerebrum and diencephalon; the rest attach to the brainstem.

THE VOCABULARY OF THE CRANIAL NERVES

Structure	What it is
Cranial nerves	Twelve pairs of nerves that arise from the brain and brainstem, part of the peripheral nervous system
Numbering	The nerves are numbered with Roman numerals I to XII, from anterior to posterior
Olfactory (I) and optic (II)	The only cranial nerves that attach above the brainstem, to the cerebrum and the diencephalon
Brainstem nerves	Cranial nerves III through XII all attach to the brainstem, where their nuclei sit
Nucleus	The cluster of cell bodies in the brain that a cranial nerve arises from or projects to
Foramen	The opening in the skull that a cranial nerve passes through to reach its target

Classifying the Cranial Nerves

Every cranial nerve is classed by the kind of signal it carries: sensory only, motor only, or both. A nerve carrying both is called mixed.

THE THREE CLASSES AND THE KINDS OF SIGNAL

Structure	What it is
Sensory nerves	Carry only sensory signals into the brain; CN I, CN II, and CN VIII
Motor nerves	Carry mainly motor signals out to muscles; CN III, CN IV, CN VI, CN XI, and CN XII
Mixed nerves	Carry both sensory and motor signals; CN V, CN VII, CN IX, and CN X
Special sensory	The senses of smell, vision, taste, hearing, and balance
Somatic motor	Voluntary control of skeletal muscle
Visceral motor	Parasympathetic control of smooth muscle and glands; carried by CN III, VII, IX, and X

A memory aid for the order of the names is the sentence "On Old Olympus' Towering Tops A Finn And German Viewed Some Hops." For the sensory, motor, or both pattern, "Some Say Marry Money But My Brother Says Big Brains Matter Most."

The Twelve Cranial Nerves

The twelve nerves in order, with the classification, the main function, and the skull foramen each one passes through. Compare them.

NUMBER, NAME, CLASS, FUNCTION, AND FORAMEN

Number	Name	Class	Primary function	Skull foramen
CN I	Olfactory	Sensory	Smell	Cribriform plate of the ethmoid
CN II	Optic	Sensory	Vision	Optic canal of the sphenoid
CN III	Oculomotor	Motor	Most eye movements, raising the eyelid, and constricting the pupil	Superior orbital fissure
CN IV	Trochlear	Motor	Eye movement by the superior oblique muscle; the smallest cranial nerve	Superior orbital fissure
CN V	Trigeminal	Mixed	Sensation from the face and the muscles of chewing; the largest cranial nerve, with three branches	Superior orbital fissure (V1), foramen rotundum (V2), and foramen ovale (V3)
CN VI	Abducens	Motor	Eye movement by the lateral rectus muscle	Superior orbital fissure
CN VII	Facial	Mixed	Facial expression, taste from the front of the tongue, and tear and saliva production	Internal acoustic meatus, then the stylomastoid foramen
CN VIII	Vestibulocochlear	Sensory	Hearing and balance	Internal acoustic meatus
CN IX	Glossopharyngeal	Mixed	Taste from the back of the tongue, swallowing, and saliva from the parotid gland	Jugular foramen
CN X	Vagus	Mixed	Autonomic control of the thoracic and abdominal organs, plus voice and swallowing	Jugular foramen

Number	Name	Class	Primary function	Skull foramen
CN XI	Accessory	Motor	Movement of the sternocleidomastoid and trapezius muscles	Jugular foramen
CN XII	Hypoglossal	Motor	Movement of the tongue	Hypoglossal canal

HOLD ONTO THIS

Three foramina carry more than one nerve. The superior orbital fissure takes III, IV, V1, and VI, the internal acoustic meatus takes VII and VIII, and the jugular foramen takes IX, X, and XI. Learn the openings and the groupings come with them.

Cranial Nerve Nuclei

Each cranial nerve connects to one or more nuclei, the clusters of cell bodies in the brain that the nerve arises from or carries signals to. Most sit in the brainstem.

WHERE EACH NERVE BEGINS OR ENDS IN THE BRAIN

Cranial nerve	Nuclei or origin
CN I, Olfactory	The olfactory epithelium projects to the olfactory bulb, then along the olfactory tract to the cortex
CN II, Optic	The retina projects through the optic nerve and the optic chiasm to the lateral geniculate nucleus
CN III, Oculomotor	The oculomotor nucleus and the Edinger-Westphal nucleus of the midbrain
CN IV, Trochlear	The trochlear nucleus of the midbrain
CN V, Trigeminal	The trigeminal ganglion and the trigeminal nuclei of the pons
CN VI, Abducens	The abducens nucleus of the pons
CN VII, Facial	The facial motor nucleus and the solitary nucleus of the pons
CN VIII, Vestibulocochlear	The cochlear and vestibular nuclei at the pons and medulla
CN IX, Glossopharyngeal	The solitary nucleus, the nucleus ambiguus, and the inferior salivatory nucleus of the medulla
CN X, Vagus	The dorsal motor nucleus, the nucleus ambiguus, and the solitary nucleus of the medulla
CN XI, Accessory	The spinal accessory nucleus of the medulla and the upper spinal cord, C1 to C5
CN XII, Hypoglossal	The hypoglossal nucleus of the medulla

TWELVE NERVES, TWELVE EXAMS

Because each cranial nerve has one job and one path, testing the nerves one by one is a fast way to localize a problem in the brainstem. A clinician who finds **CN VI** weak knows to look at the pons, where its nucleus sits.

The foramina group nerves together. The **jugular foramen** carries CN IX, X, and XI side by side, so a tumor or fracture there can knock out all three at once, producing a recognizable cluster of swallowing, voice, and shoulder signs.

Two of the most common nerve problems are cranial. **Bell's palsy** is a sudden weakness of CN VII that drops one side of the face. **Trigeminal neuralgia** is a stabbing facial pain along the branches of CN V. The anatomy of the nerve predicts exactly where each one shows up.